Reading 6.2: MDP and Optimal Policy

Welcome to our session on Markov Decision Processes (MDPs), the Bellman Equations and the Optimal Policy. These elements are the foundation of sequential decision-making. By the end of this session, you will:

- Understand the components of an MDP
- Learn how the 4 Bellman equations help calculate state values V(s) and state-action values Q(s,a)
- Grasp the concept of an Optimal Policy π_*

The Markov Decision Process formally describes the dynamics of the environment, assuming that the environment is fully understood by the agent. MDPs are a classical formalization of sequential decision-making problems where actions influence immediate and future rewards. MDPs have been studied extensively since the 1950s. We will explain how an MDP works relying on the dynamics of a small ski resort. An MDP is defined by 5 parameters:

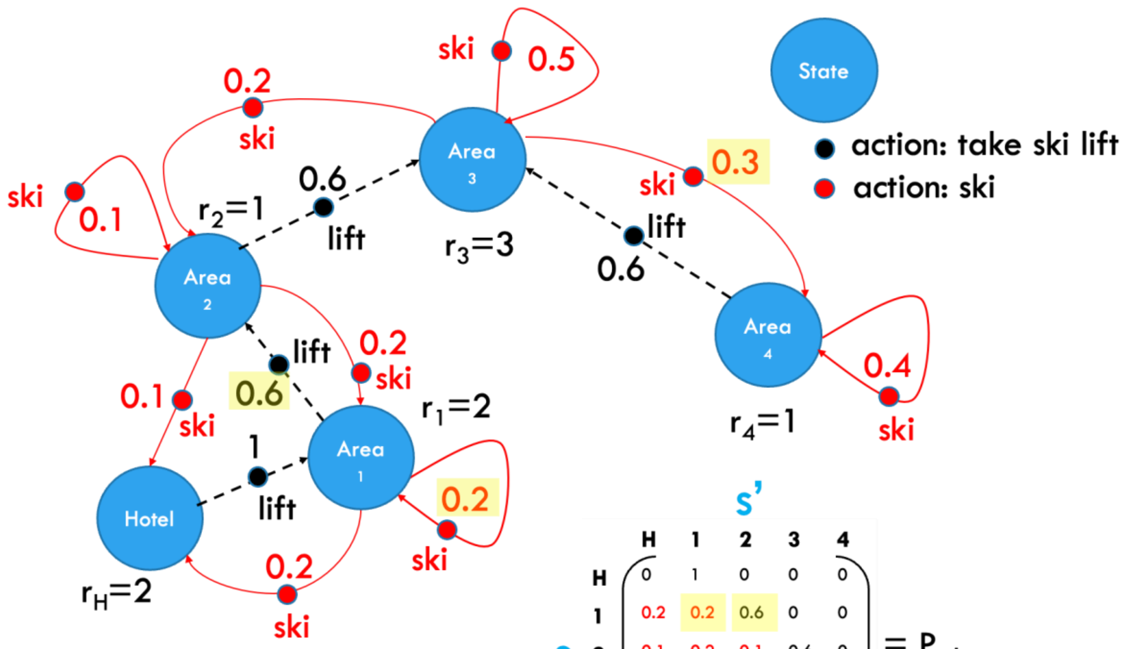
1. a set of states – we have 5 states: Hotel, Ski Area1, Ski Area2, Ski Area3 and Ski Area4
2. a set of actions – an action is anything that an agent can do while in a specific state: taking a lift, ski in the Area or ski to another Area
3. a set of rewards – an agent gets a reward whenever it observes one of the states
4. state transition probabilities (Pss') – probabilities of moving from a state s to the next state s'
5. discount factor γ – allows us to manage the relative importance of immediate and future rewards

Instead of using st and st+1 to represent the current and next state, we will often use s and s' as well.

The solution of an MDP is a policy π. The policy can be thought of as a table linking states to actions, representing what action the agent should take in a given state.

In the morning all skiers take the lift near the hotel to Area1. People arriving in Area1, have the option to ski in Area1 with probability 20% (0.2), take the lift to Area2 with probability 60% (0.6) or ski back to the hotel with probability 20% (0.2).

Once people arrive in Area2, they will either stay in this area with probability 10%, take the ski lift to Area3 with probability 60%, ski back to Area1 with probability 20% or ski back to the hotel with probability 10%. We can describe similar dynamics for Area3 and Area4. At the end of the day all skiers are back in the hotel, our terminal state.



$$
\begin{array}{c|c}
 & \begin{array}{ccccc} H & 1 & 2 & 3 & 4 \end{array} \\
\begin{array}{c} H \\ 1 \\ 2 \\ 3 \\ 4 \end{array} &
\begin{pmatrix}
0 & 1 & 0 & 0 & 0 \\
0.2 & 0.2 & 0.6 & 0 & 0 \\
0.1 & 0.2 & 0.1 & 0.6 & 0 \\
0 & 0 & 0.2 & 0.5 & 0.3 \\
0 & 0 & 0 & 0.6 & 0.4
\end{pmatrix} = P_{ss'}
\end{array}
$$

The dynamic process of skiers going from one area/state (s) to another area/state (s') is represented by the transition probability matrix Pss'. For example, the probability to transition from Area1 to Area2 is 60%. The probability to stay in Area1 is 20%. The probability to go from Area3 to Area4 is 30%. The rewards ri are banked immediately upon arrival in the successor state.

Bellman Equations

The Bellman equation expresses the relationship between the value V_π(s) of the current state and the value of all potential successor states V_π(s'). The return G of the current state s is defined as follows:

$$G(s) = \overbrace{r'}^{\text{immediate reward}} + \underbrace{\gamma r'' + \gamma^2 r''' + \ldots}_{\text{discounted future rewards}}$$

Simple algebra allows us to express the return of the current state G(s) as a function of the return of the next state G(s').

$$G(s) = r' + \gamma(\underbrace{r'' + \gamma r''' + \ldots.}_{G(s')})$$

$$G(s) = r' + \gamma G(s') \quad (1)$$

We also defined the state value V_π(s) as the expected value of the return G(s)

$$\text{relying on (1)} \quad V_\pi(s) = E_\pi\,[\mathrm{r'} + \gamma G(\mathrm{s'})]$$

$$V_\pi(s) = \mathrm{r'} + \gamma E_\pi[G(\mathrm{s'})]$$

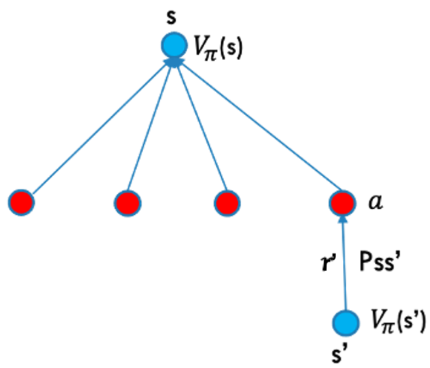
$$\boxed{V_\pi(s) = \mathrm{r'} + \gamma V_\pi(s')} \quad (2)$$

Let's generalize equation (2) and derive the Bellman equation for an MDP with multiple actions and successor states.

$$\boxed{V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} \sum_{r'} P_{ss'}[\mathrm{r'} + \gamma V_\pi(\mathrm{s'})]}$$

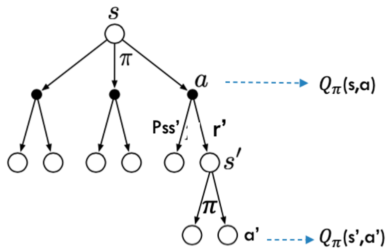
(Bellman 1)

Bellman Backup Diagram



This Bellman equation links the value of a state s to the value of its potential successor states s' without the need to observe all future rewards. Linking V(s) and V(s') via Bellman provides us with an algorithm to calculate state values given a policy π. We work our way backwards and determine $V_\pi(s)$ by considering all actions that the agent might take and dealing with everything that the environment might throw at us in the next state $V_\pi(s')$. Determining $V_\pi(s)$ via an estimate of $V_\pi(s')$ is called bootstrapping. The associated diagram is called the Bellman backup diagram.

Just as there is a Bellman equation for state values, there is also a Bellman equation for state-action:



$$Q_\pi(s,a) = E_\pi\,[G_t\,|\,s,a]$$

$$\boxed{Q_\pi(s,a) = \sum_{s'}\sum_{r'} P_{ss'}[r' + \gamma \sum_{a'} \pi(a'|s')\, Q_\pi(s',a')]}$$ **(Bellman 2)**

Optimal Policy π_*

In Reinforcement Learning, we are interested in finding the optimal policy π_* or the policy that maximizes the agent's cumulative rewards over time, in expectation. It can be proven that, for every MDP, there exists always an optimal deterministic policy π_*. The optimal state value and state-action value are linked to the optimal policy via:

$$V_{\pi*}(s) = E_{\pi*}[G_t(s)] = max_\pi\, V_\pi(s)$$

$$Q_{\pi*} = max_\pi\, Q_\pi(s, a)$$

We can re-write the Bellman equation for the optimal values:

$$\boxed{V_{\pi_*}(s) = max_a\, [\sum_{s'}\sum_{r'} P_{ss'}[r' + \gamma V_{\pi_*}(s')]}$$ **(Bellman 3)**

$$\boxed{Q_{\pi_*}(s,a) = \sum_{s'}\sum_{r'} P_{ss'}[r' + \gamma max_{a'} Q_{\pi_*}(s',a')]}$$ **(Bellman 4)**

Try it Yourself

Note that the $\max_{a'}$ term is non-linear, and we cannot solve this equation with our linear algebra machinery.

All 4 Bellman equations assume that we have access to Pss' or the transition probability matrix. In many real-life situations this matrix in unknown. Think about how this can be solved.